

QueryBook — Drawings, Reference Numerals & Brief Description of the Drawings

Scope: the complete, current QueryBook figure set — FIG. 1–FIG. 34 — as one unified set: the integral language & speech subsystem (LEL; FIG. 1–8, 31–32), the QueryBook platform (FIG. 9–30), and the Language & Identity Linking (LIL) voiceprint & identity subsystem (FIG. 33–34). **Companion** to the consolidated *QueryBook Provisional Patent Application*; the figures, reference numerals, and brief descriptions here are identical to those in that application. **Document date:** October 1, 2026. Draft — to be finalized to USPTO drawing standards (37 CFR 1.84) by a patent draftsman before filing.

Brief Description of the Drawings

FIG. 1 is a block diagram of the overall QueryBook Language Expression Layer (LEL) system architecture.

FIG. 2 is a block diagram of the Sub-Language Priming Layer (SLPL) and the measurable representation produced by each of its six components.

FIG. 3 is a flow diagram of the developmental learning sequence and the one-way Transition Gate having four readiness thresholds.

FIG. 4 is a block diagram of the multilingual Delta Acquisition Model and the acquisition acceleration flywheel.

FIG. 5 is a block diagram of the seven-stage Speech Output Pipeline.

FIG. 6 is a data-flow diagram of the co-grounded integration of the LEL with the Federated Query Language (FQL) and the Universal Federated Connectivity System (UFCS).

FIG. 7 is a block diagram of a reduced-to-practice embodiment (the Language Lab) implementing Phase One, the Transition Gate, and a deterministic Phase Two, per Section VII.

FIG. 8 is a data-flow diagram of the deterministic Phase-Two semantic grounding of the reduced-to-practice embodiment, in which word meanings are attached only from auditable sources — a bundled public-domain dictionary and self-grounding against verified Fact Units in the store — with any large-language-model meaning source excluded from the grounding path.

Brief Description of the QueryBook Platform Drawings (FIG. 9–FIG. 30)

FIG. 9 is a diagram of the queryBook system architecture (Prime Directive as Domain 0).

FIG. 10 is a diagram of the fact Unit — atom, envelope, and content-derived identity.

FIG. 11 is a diagram of the fact-certification authority hierarchy.

FIG. 12 is a diagram of the uFCS substrate — three-store architecture.

FIG. 13 is a diagram of the authored tour and exhibition sequencing.

FIG. 14 is a diagram of the fQL — joint structure and trust execution.

FIG. 15 is a diagram of the verifying machine-generated text against certified records.

FIG. 16 is a diagram of the federation — cross-boundary re-validation.

FIG. 17 is a diagram of the fact-level erasure by external-key destruction.

FIG. 18 is a diagram of the universal security architecture.

FIG. 19 is a diagram of the universal honeypot saturation and data firehose.

FIG. 20 is a diagram of the deployment topologies over an invariant common layer.

FIG. 21 is a diagram of the adaptive response engine — directive fixed before generation.

FIG. 22 is a diagram of the user-owned agent system with governance and kill switch.

FIG. 23 is a diagram of the deterministic multi-language regeneration.

FIG. 24 is a diagram of the semantic compression, expansion, colorization and upscaling.

FIG. 25 is a diagram of the queryBook-enhanced TCP/IP hybrid protocol packet.

FIG. 26 is a diagram of the neuromorphic cognitive layer — attractor convergence.

FIG. 27 is a diagram of the fact Unit — confidence and evidence model.

FIG. 28 is a diagram of the uFCS packet structure — serialized Fact Unit record.

FIG. 29 is a diagram of the uFCS packet framing — content-addressed block storage.

FIG. 30 is a diagram of the fQL — query anatomy and worked example.

Reference Numerals

No.	Element	No.	Element
100	QueryBook LEL system	732	rule-seeded G2P
110	Sub-Language Priming Layer (SLPL)	733	syllabifier
120	Language Expression Layer (LEL)	734	lexical-stability metric
130	Transition Gate	735	co-occurrence density metric
140	Phase Two semantic-communicative learning	736	prosody proxy
150	Multilingual Delta Acquisition module	740	Fact Unit store
160	Speech Output Pipeline	750	Transition-Gate metrics
165	natural speech output	751	one-way latch
170	co-grounding interface	760	phase agents on shared manager
180	Federated Query Language (FQL)	761	Phase-1 agent (implemented)
190	Universal Federated Connectivity System (UFCS)	762	Phase-2 agent — deterministic dictionary + store grounding (implemented)
195	federated data sources	763	Phase-3 agent (declines — spec-only)
200	SLPL	764	Phase-4 agent (declines — spec-only)
210	acoustic phonetic processing	800	deterministic Phase-Two semantic grounding
211	signal-level feature embedding	810	Phase-One vocabulary (learned words)
220	phonics / phonemic awareness	820	deterministic grounder (per word)
221	phoneme-boundary segmenter	830	bundled public-domain dictionary source
230	communication-science foundations	831	dictionary meaning Fact Unit (means, trust 0.9)
231	communicative-salience weight	840	self-grounding store-lookup source
240	prosody & rhythm processing	841	store-link Fact Unit (grounded_by_fact)
241	prosodic contour vector	850	Fact Unit store (provenance, domain language)
250	information-theoretic grounding	860	excluded large-language-model meaning source
251	information weighting	870	optional future dictionary-gated suggestor
260	neurolinguistic chunking	900	advanced multilingual pronunciation subsystem
261	chunk segments	910	pronunciation request (word/phrase, language)
270	licensed multimedia input	920	pronunciation preference-ladder selector
300	developmental learning sequence	930	bundled open-source phonemizer
310	Phase One (structural-acoustic)	931	phonemizer IPA transcription
311	sound discrimination	932	phonemizer call cache
312	syllabic segmentation	933	bounded call log / timeout guard
313	word-as-acoustic-object	940	rule-seeded G2P tables (per language)
314	semantics suppressed	941	approximate rule-seeded IPA (offline)
320	Transition Gate (one-way)	950	syllable count + primary-stress derivation
321	phonemic-inventory completeness $\geq \tau_1$	960	explicit null result (not fabricated)
322	lexical stability $\geq \tau_2$	970	phrase-level single-call transcription
323	co-occurrence density $\geq \tau_3$	971	displayed transcription consistent with synthesized audio
324	pattern saturation $< \epsilon$	980	self-contained application bundle
330	Phase Two (semantic-communicative)	981	bundled open-source speech engine
331	word-to-referent grounding	982	runtime-support libraries
332	communicative discovery	983	bundled phoneme data
333	emergent grammar / syntax	984	engine-selection ladder (per OS / language)
400	Delta Acquisition Model	985	host OS built-in voice service
410	reused L1 foundation	986	culture-matched OS voice (when installed)
420	L2 delta detector	987	honest refusal with remedy (no fabricated audio)
430	targeted training of novel elements	990	deterministic translator (through English)
440	multilingual semantic alignment	991	bundled open-source bilingual dictionary
450	acquisition acceleration flywheel	992	coverage accounting + unknown-word passthrough (flagged)
451	negative-transfer detection & retraining	993	deterministic language detector
500	Speech Output Pipeline	995	permanent quality-control battery
510	lexical selection	996	IPA-produced check
520	grapheme-to-phoneme conversion	997	syllable-sanity check (± 1)
530	coarticulation modeling	998	valid-waveform synthesis check
540	prosody generation	999	per-language QC report + overall pass score
550	register & affect calibration	1000	LIL voiceprint & identity subsystem
560	self-monitoring feedback loop	1001	consent gate (precondition)
570	neural vocoder rendering	1002	audio/video capture + VAD/segmentation
580	natural speech waveform	1003	acoustic feature extraction

No.	Element	No.	Element
600	co-grounded integration	1004	speaker embedding (voiceprint vector)
610	natural-language input	1005	enrollment — canonical voiceprint
620	LEL analysis	1006	match / score (cosine · PLDA)
630	FQL operations	1007	QueryBook trust gate (VERIFIED/CONTRADICTED/UNKNOWN)
640	UFCS execution	1008	Fact-Unit identity graph
650	federated sources	1009	LEL cross-script name/entity resolution
660	results	1010	provenance-tracked link Fact Unit
670	render to NL / speech in input language	1020	voice authentication request
680	language-agnostic resolution	1021	capture phrase + embed
700	Language Lab prototype embodiment	1022	anti-spoofing (liveness/replay/synthetic)
710	corpus source	1023	deterministic risk-scoring policy
720	Phase-One structural agent	1024	authentication decision gate
730	structural analyzer	1025	second-factor challenge
731	grapheme inventory	1026	auth-event Fact Unit (provenance)

Drawings

Formal patent drawings — black line art, forming **one QueryBook figure set**: FIG. 1–FIG. 8 and FIG. 31–FIG. 32 depict QueryBook’s integral developmental language and speech subsystem (the Language Expression Layer), FIG. 9–FIG. 30 depict the QueryBook platform in which that subsystem is co-grounded (see FIG. 6), and FIG. 33–FIG. 34 depict the Language & Identity Linking (LIL) voiceprint and multimodal identity subsystem. Each figure is on its own sheet and every element is identified by a reference numeral used consistently in the Brief Description of the Drawings, the Reference Numerals key, and the Detailed Description. Figures are draft; counsel/draftsperson to finalize to USPTO drawing standards (37 CFR 1.84) before filing.

FIG. 1

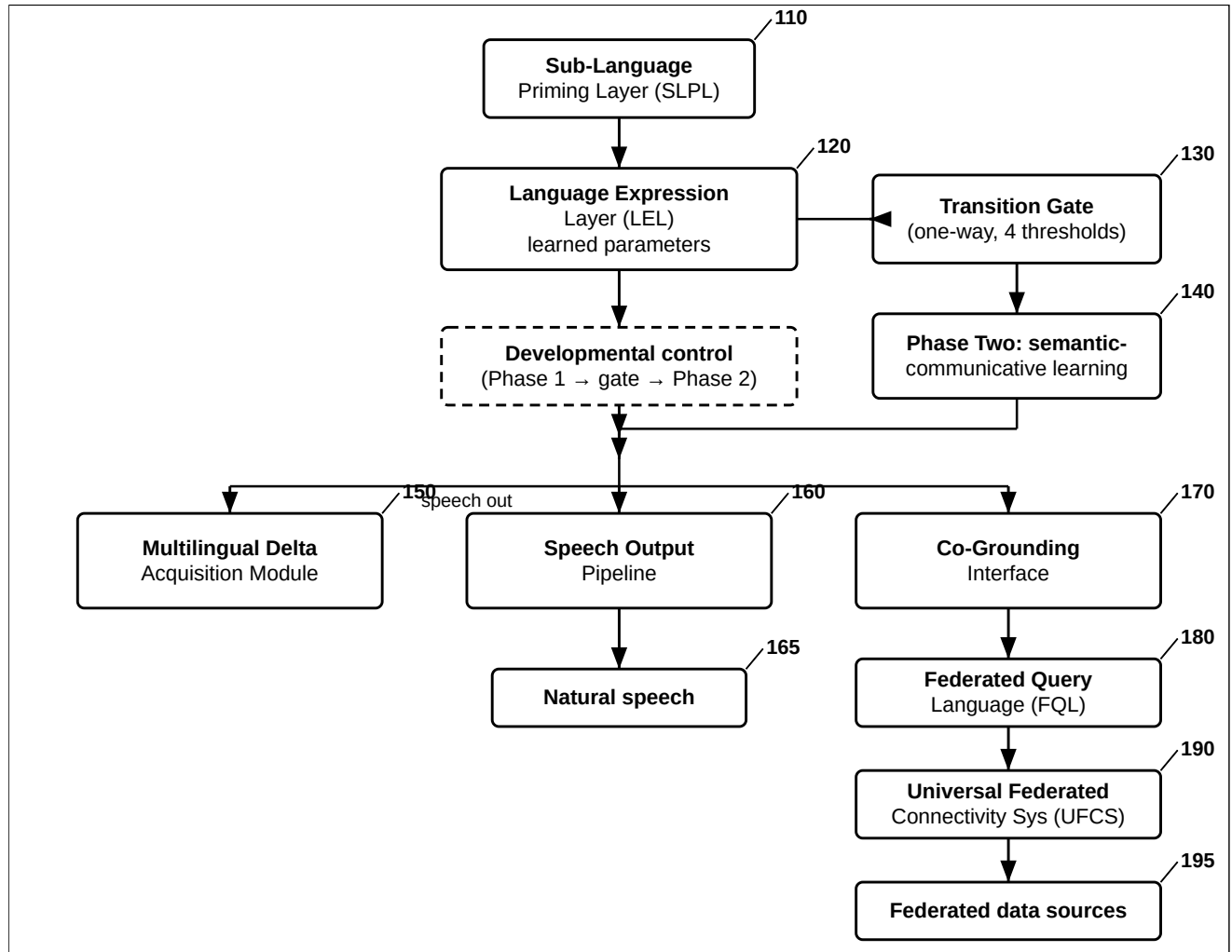


FIG. 1 — QueryBook LEL — Overall System Architecture

FIG. 2

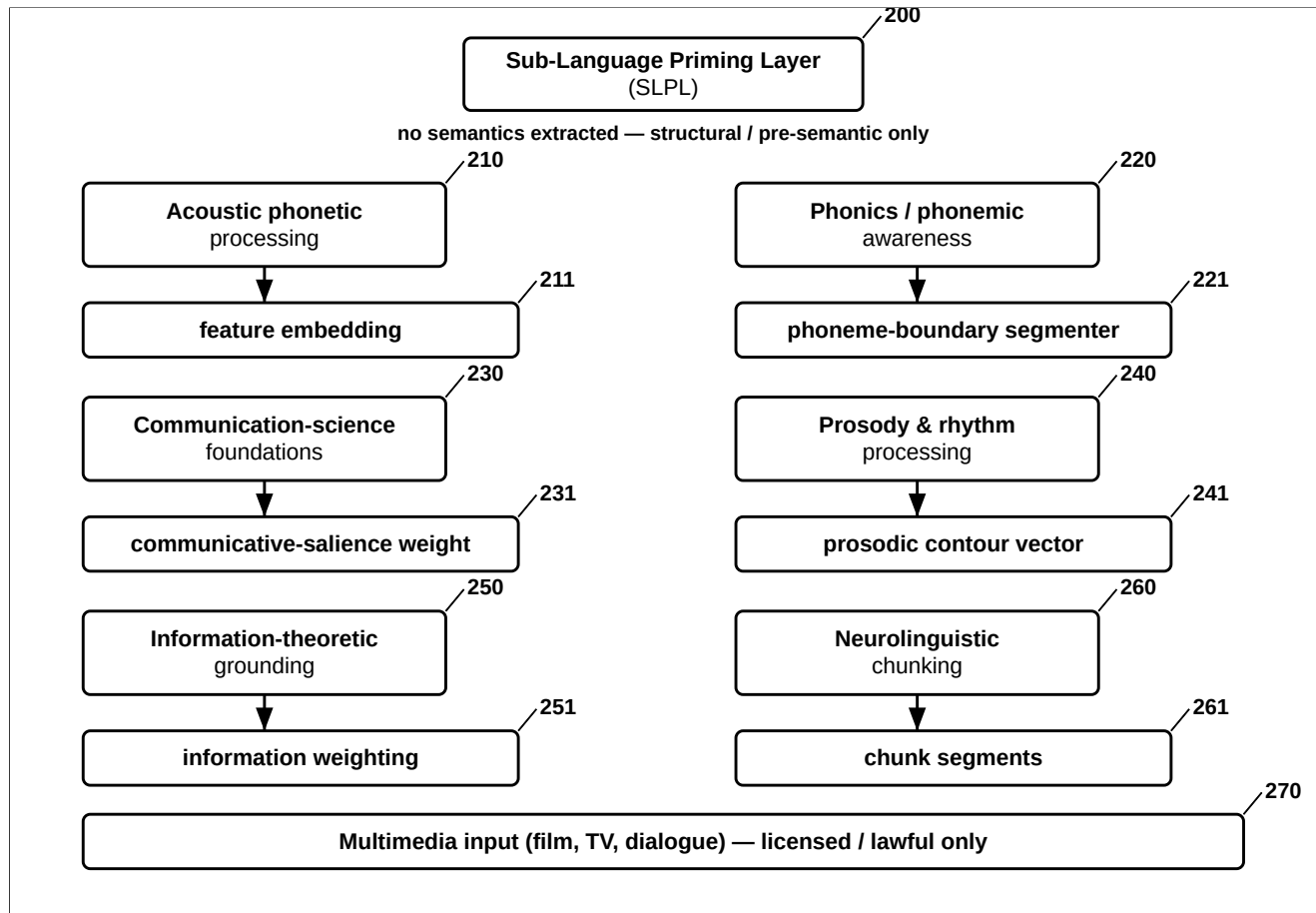


FIG. 2 — Sub-Language Priming Layer (SLPL) and Its Measurable Outputs

FIG. 3

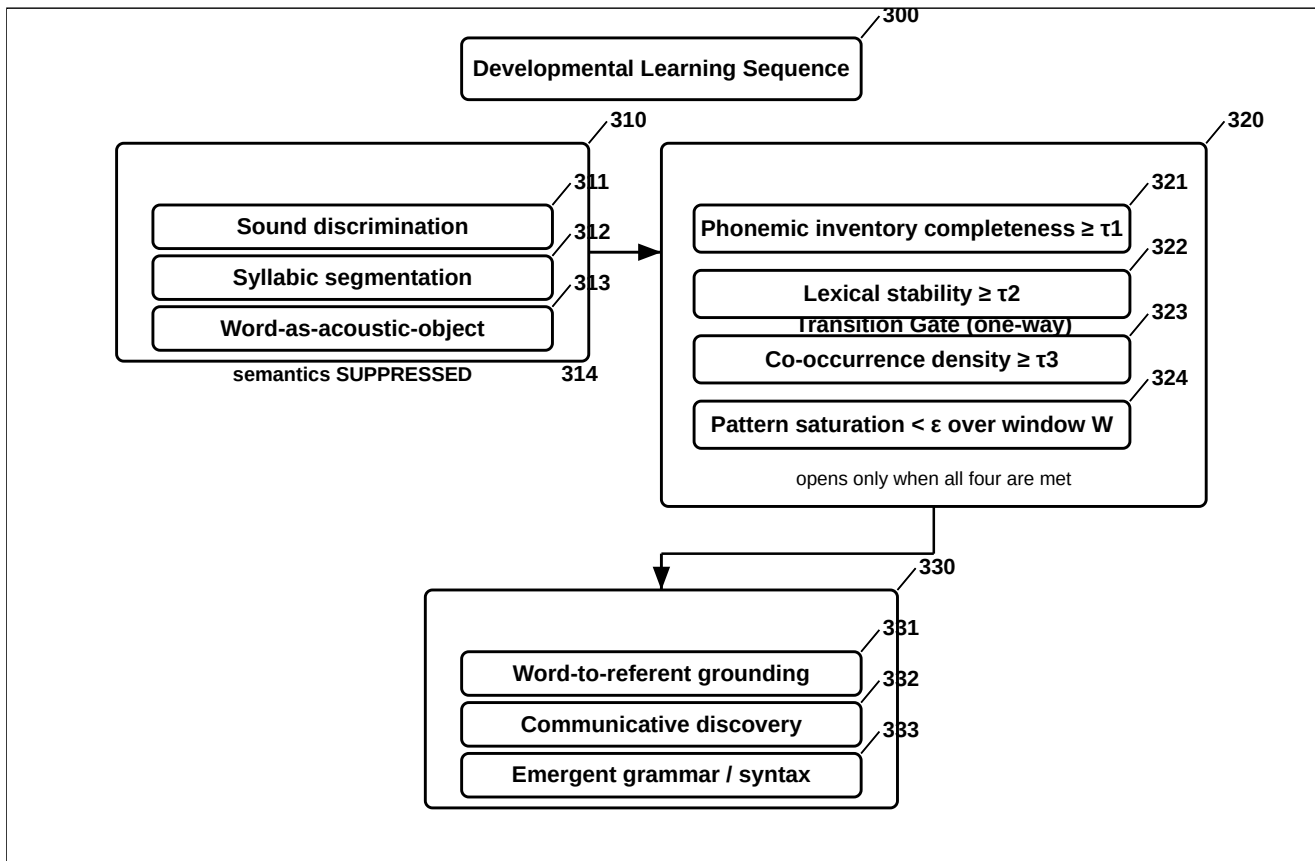


FIG. 3 — Developmental Learning Sequence and One-Way Transition Gate

FIG. 4

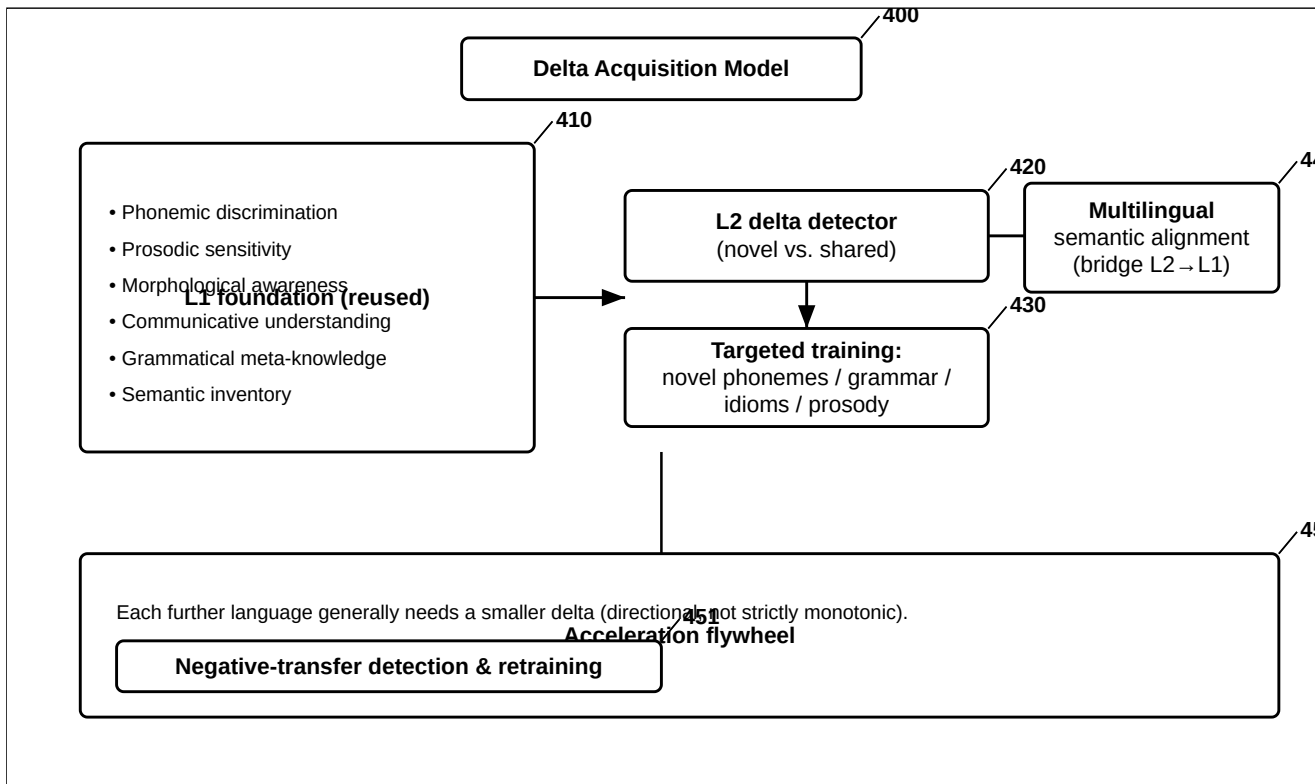


FIG. 4 — Multilingual Transfer — Delta Acquisition and Acceleration Flywheel

FIG. 5

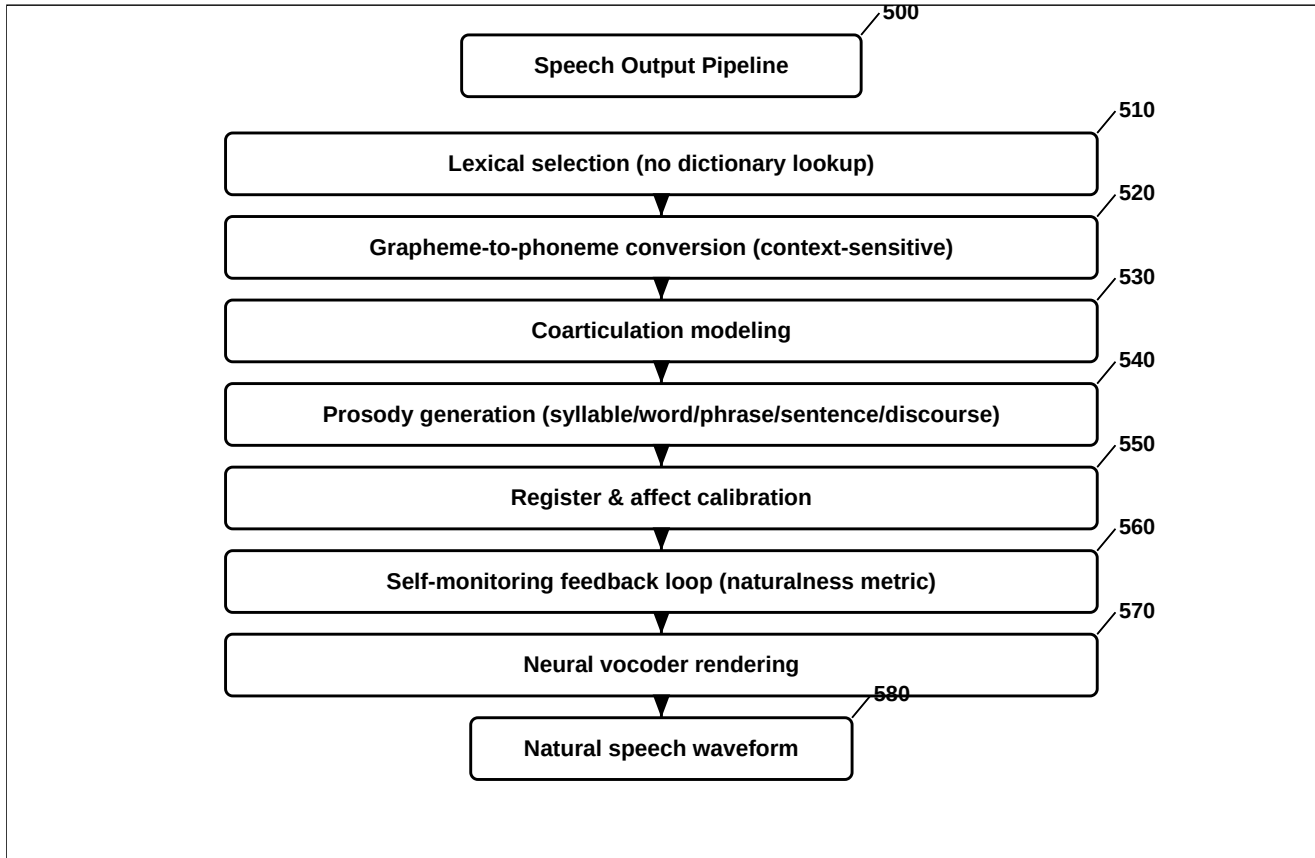


FIG. 5 — Speech Output Pipeline (Seven Learned Stages)

FIG. 6

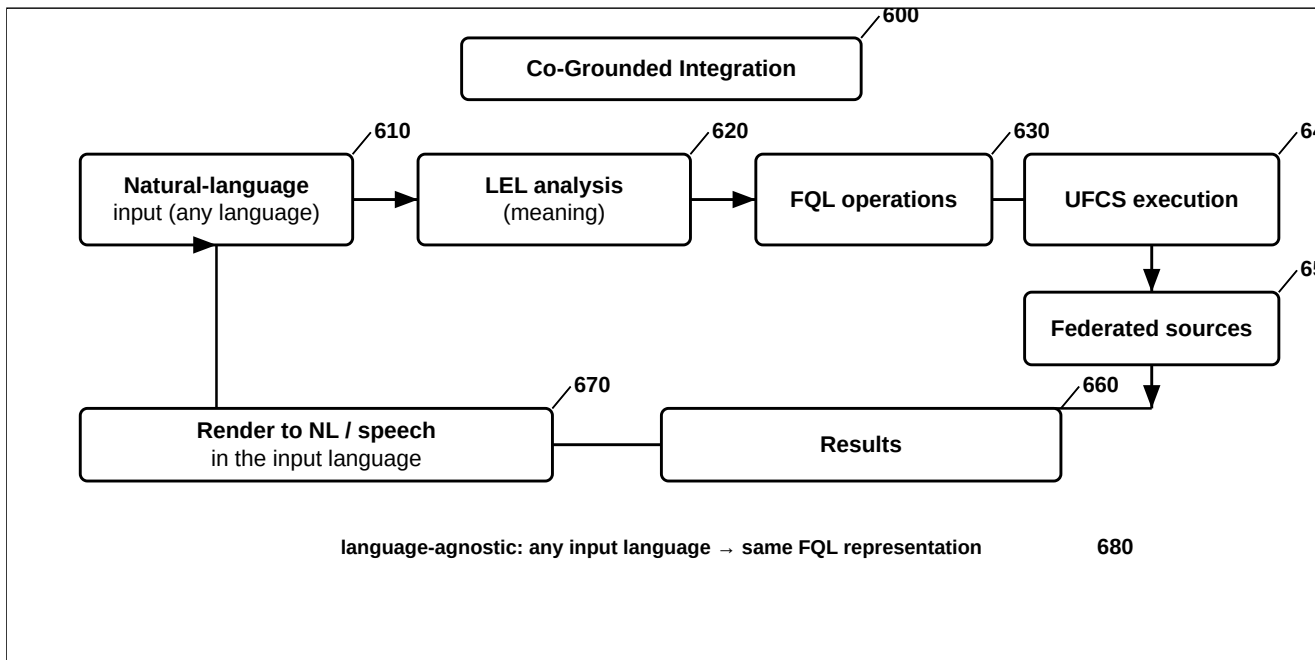


FIG. 6 — Co-Grounded Integration with FQL and UFCS

FIG. 7

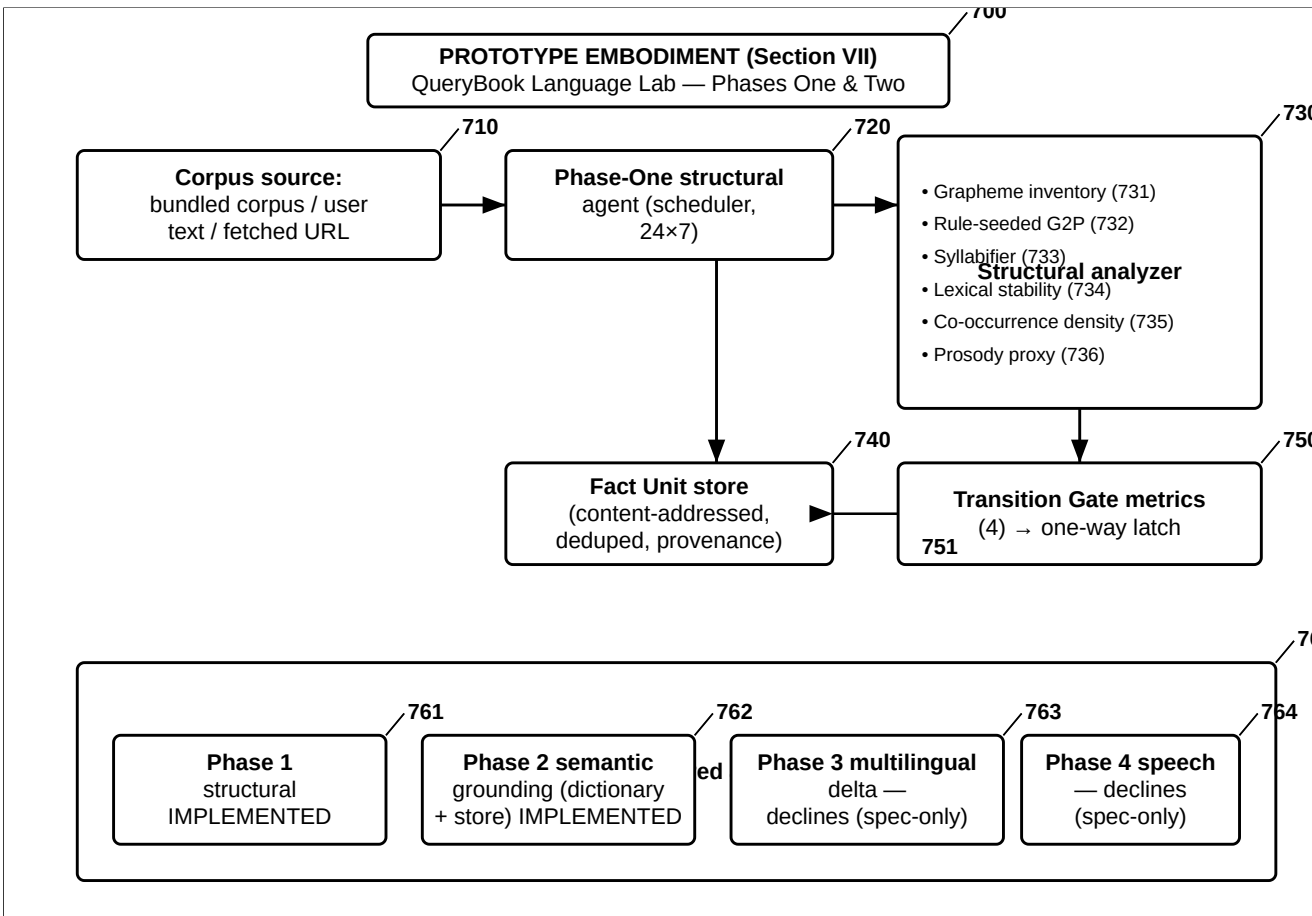


FIG. 7 — Reduced-to-Practice Embodiment — Language Lab Phases One and Two Prototype

FIG. 8

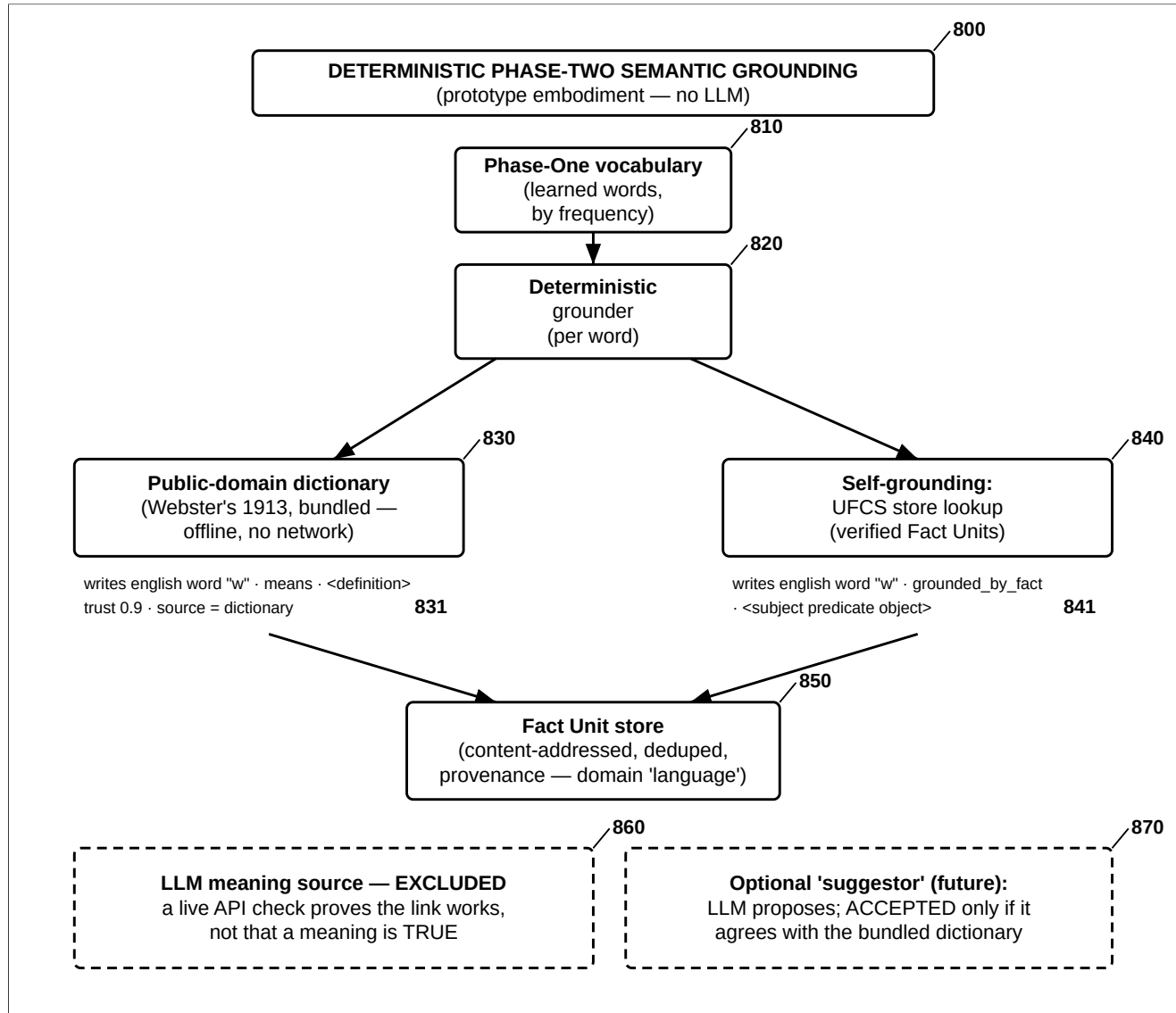


FIG. 8 — Deterministic Phase-Two Semantic Grounding — Dictionary and Store (No LLM)

QueryBook Platform Drawings (FIG. 9–FIG. 30) — the system in which the language subsystem is integral

FIG. 9 through FIG. 30 depict the QueryBook deterministic, provenance-tracked federated knowledge and query platform of which the developmental language and speech subsystem (the Language Expression Layer, FIG. 1–FIG. 8 and FIG. 31–FIG. 32) is an integral, co-grounded part (see FIG. 6). Together with the language-subsystem figures these form **one QueryBook figure set**. Draft — the language-subsystem figures use a 100–999 numeral series and the platform figures a 100–2200 series; counsel/draftsperson to consolidate the two into a single QueryBook numeral series and finalize to 37 CFR 1.84 before filing.

FIG. 9

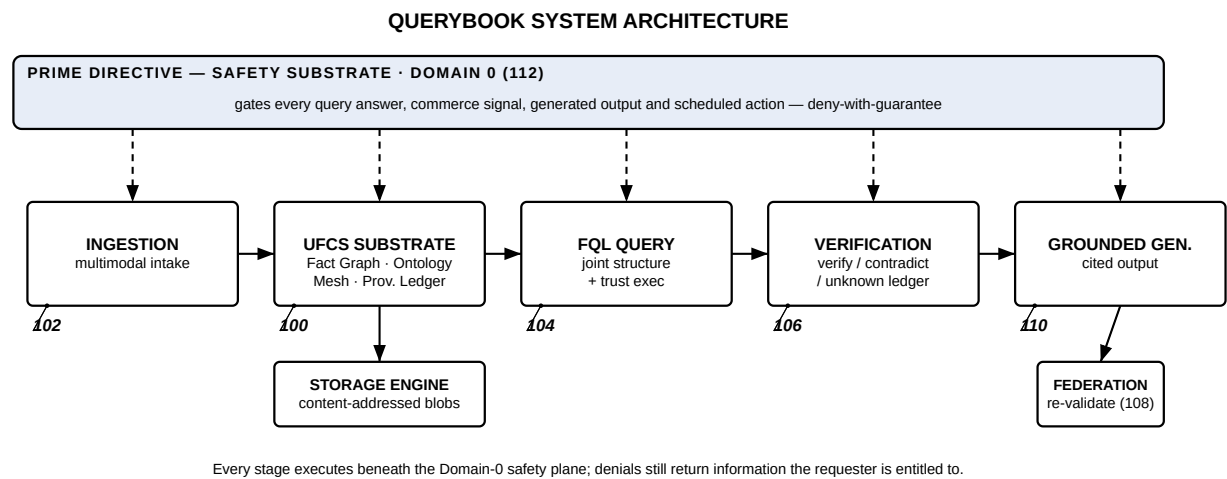


FIG. 9

FIG. 9 — QueryBook system architecture (Prime Directive as Domain 0)

FIG. 10

FACT UNIT — ATOM, ENVELOPE, AND CONTENT-DERIVED IDENTITY

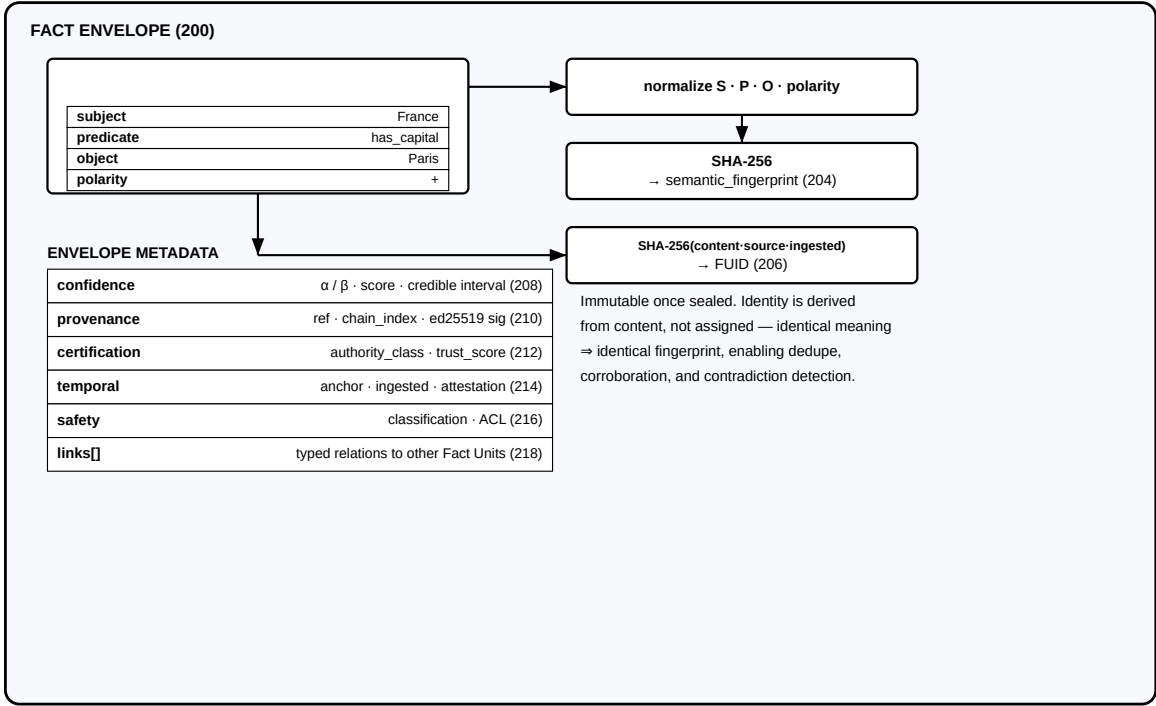


FIG. 10

FIG. 10 — Fact Unit — atom, envelope, and content-derived identity

FIG. 11

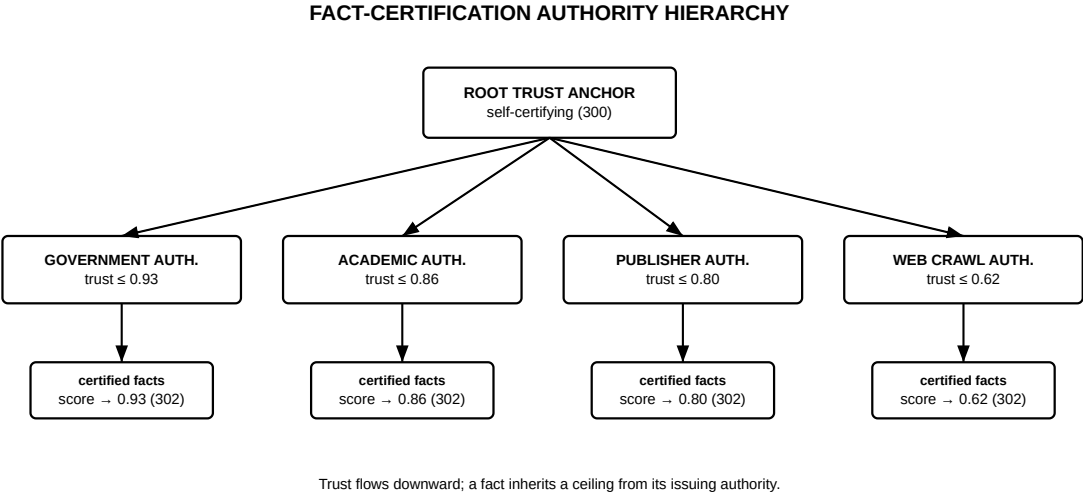


FIG. 11

FIG. 11 — Fact-certification authority hierarchy

FIG. 12

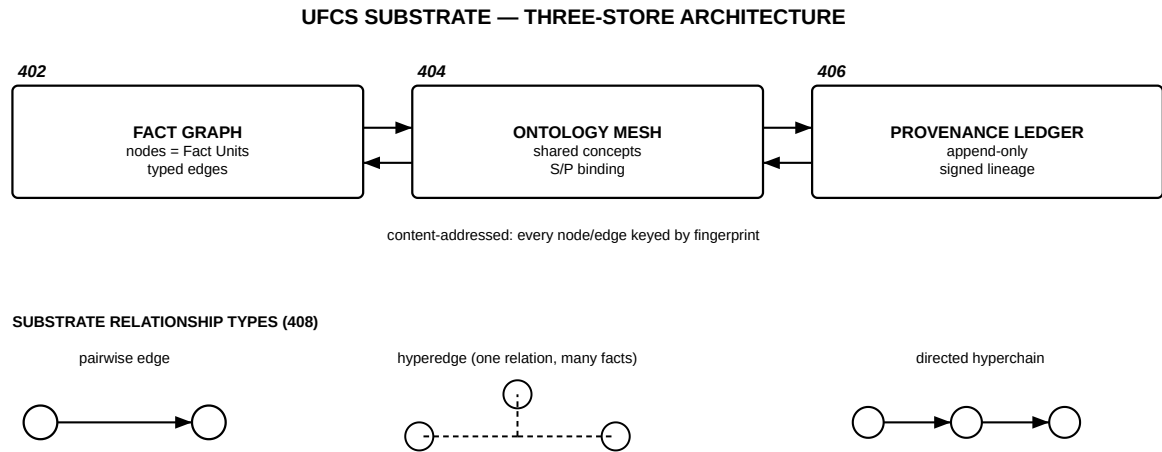
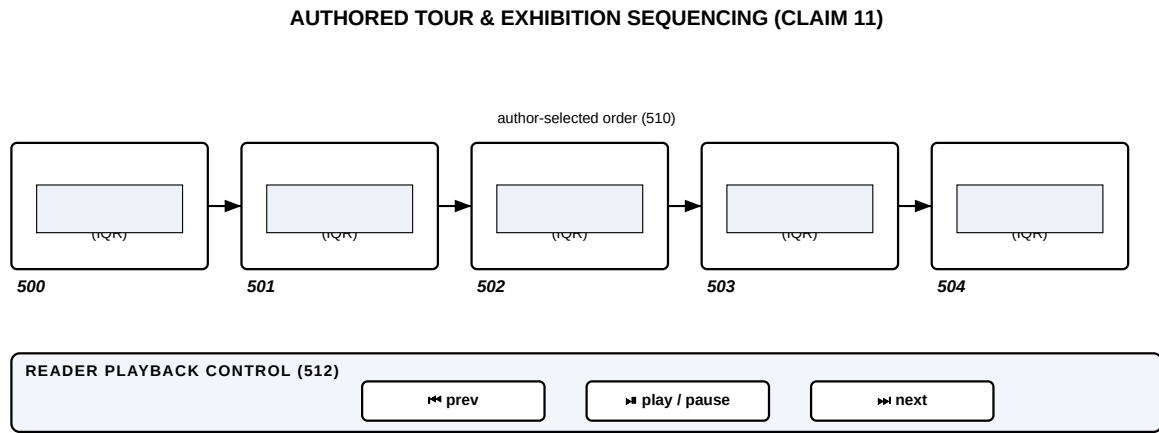


FIG. 12

FIG. 12 — UFCS substrate — three-store architecture

FIG. 13



Author fixes the sequence; the reader controls pace and direction along the ordered waypoints.

FIG. 13

FIG. 13 — Authored tour and exhibition sequencing

FIG. 14

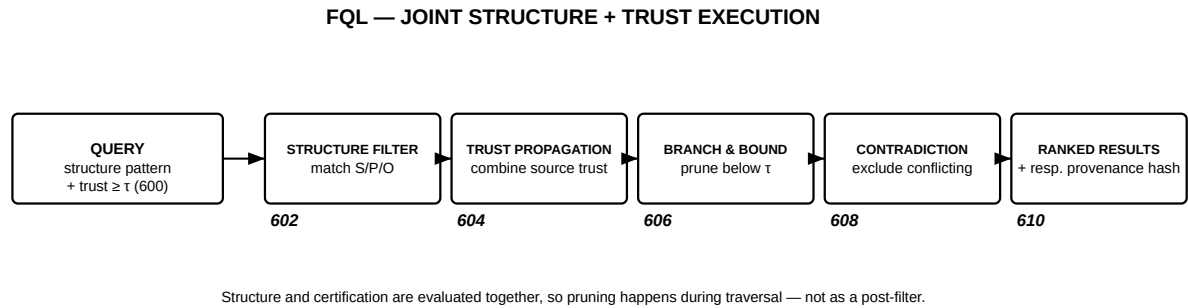


FIG. 14

FIG. 14 — FQL — joint structure and trust execution

FIG. 15

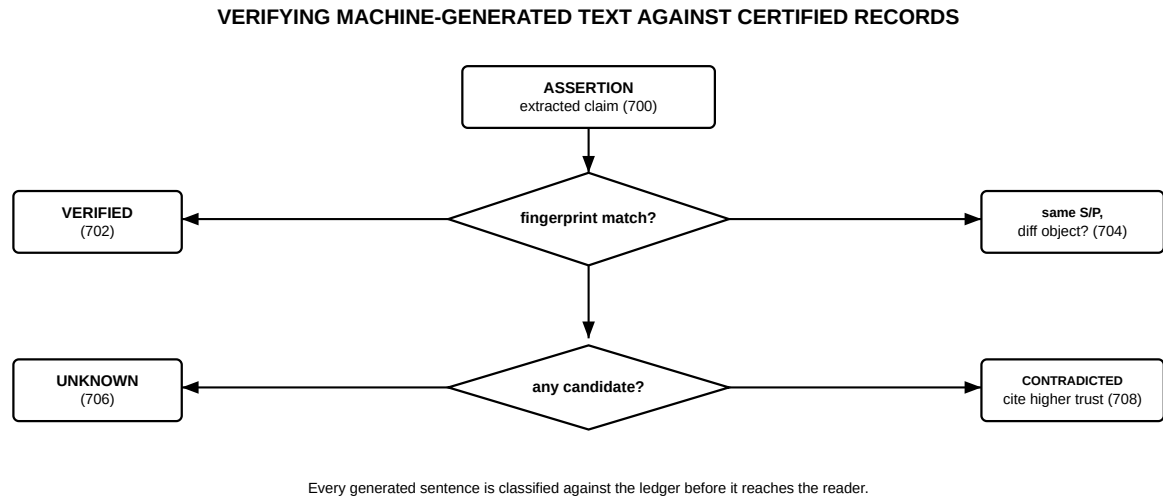


FIG. 15

FIG. 15 — Verifying machine-generated text against certified records

FIG. 16

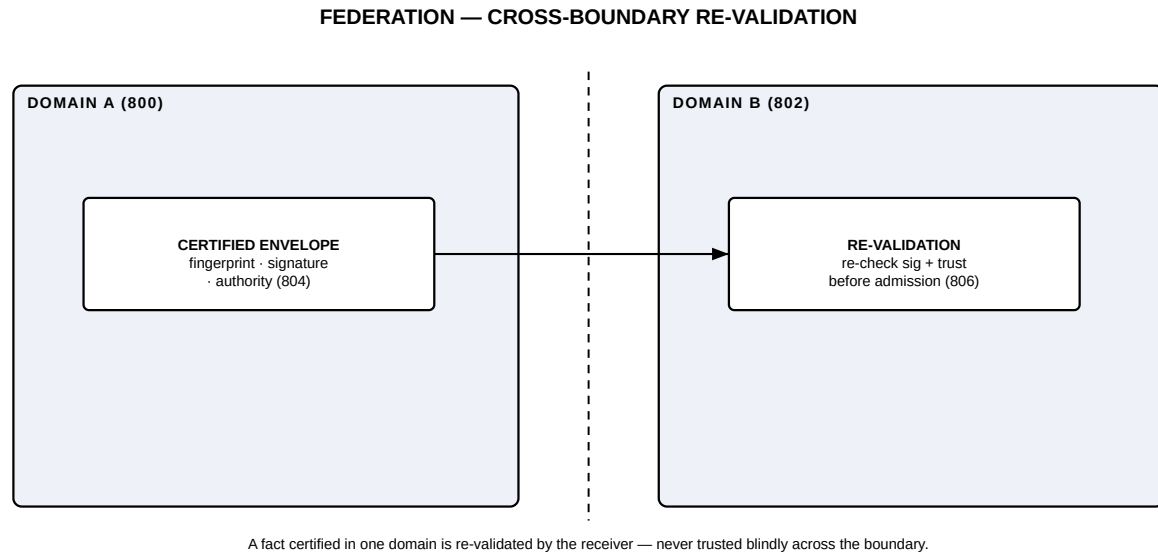
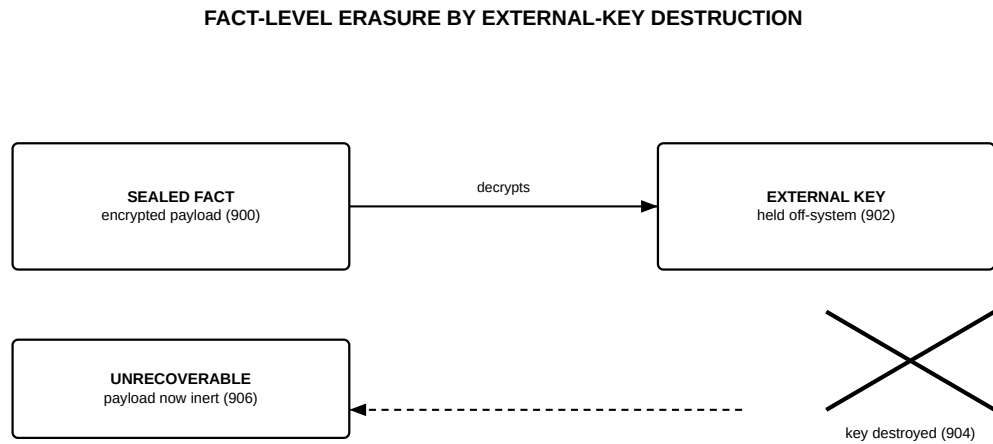


FIG. 16

FIG. 16 — Federation — cross-boundary re-validation

FIG. 17



Erase one fact by destroying its key; the content-addressed store is otherwise append-only and immutable.

FIG. 17

FIG. 17 — Fact-level erasure by external-key destruction

FIG. 18

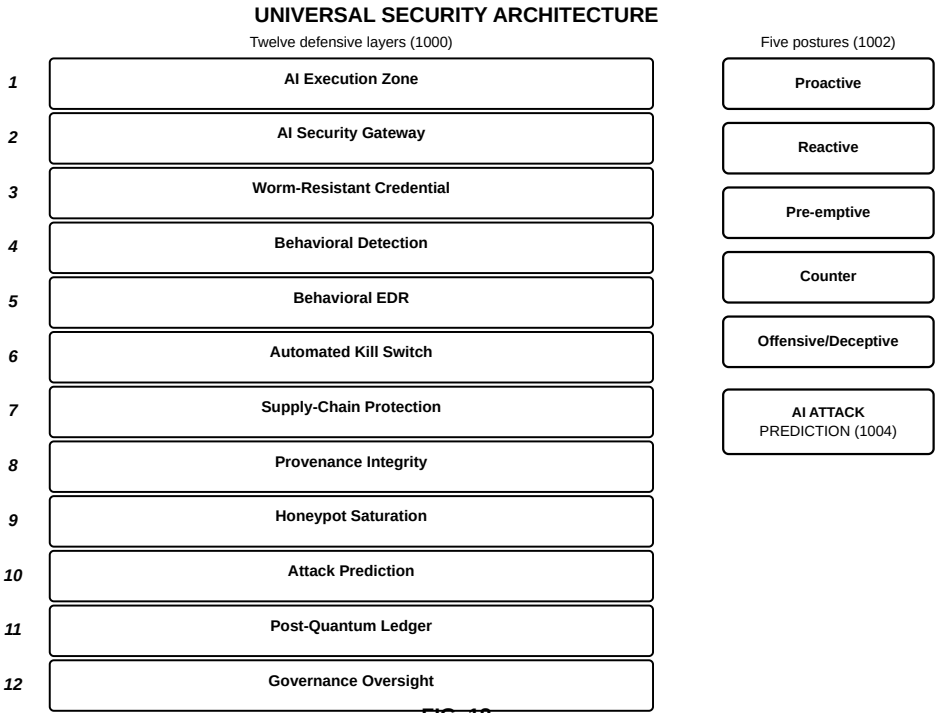


FIG. 18 — Universal security architecture

FIG. 19

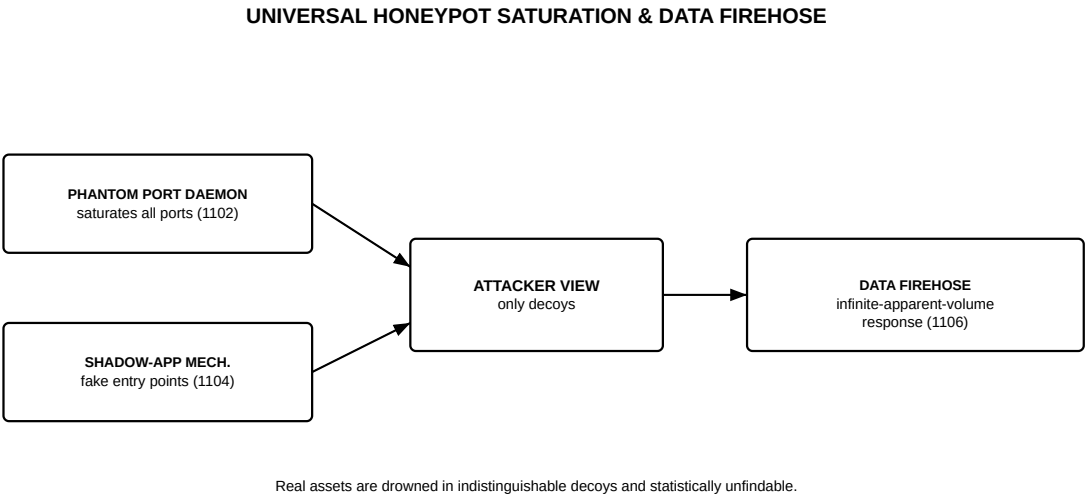


FIG. 19

FIG. 19 — Universal honeypot saturation and data firehose

FIG. 20

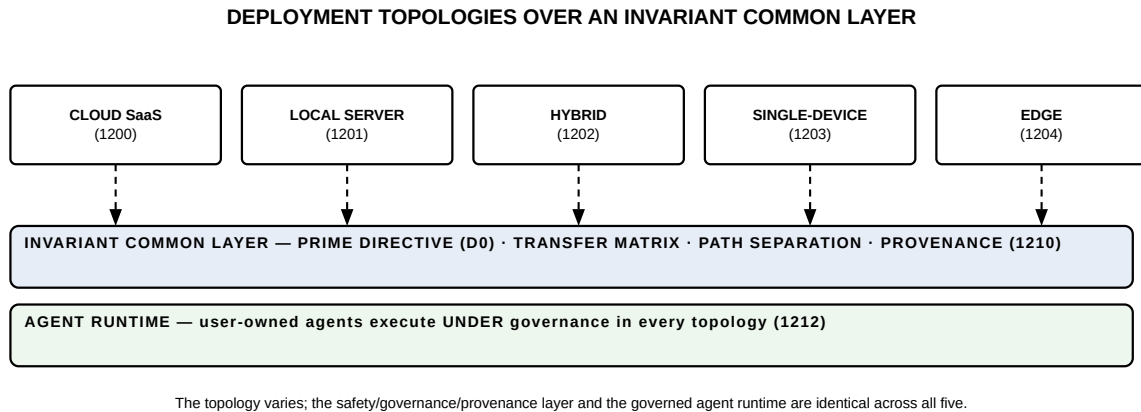


FIG. 20

FIG. 20 — Deployment topologies over an invariant common layer

FIG. 21

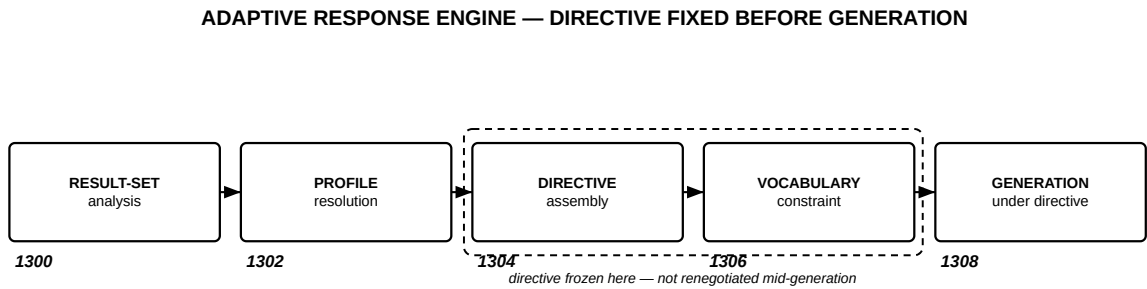
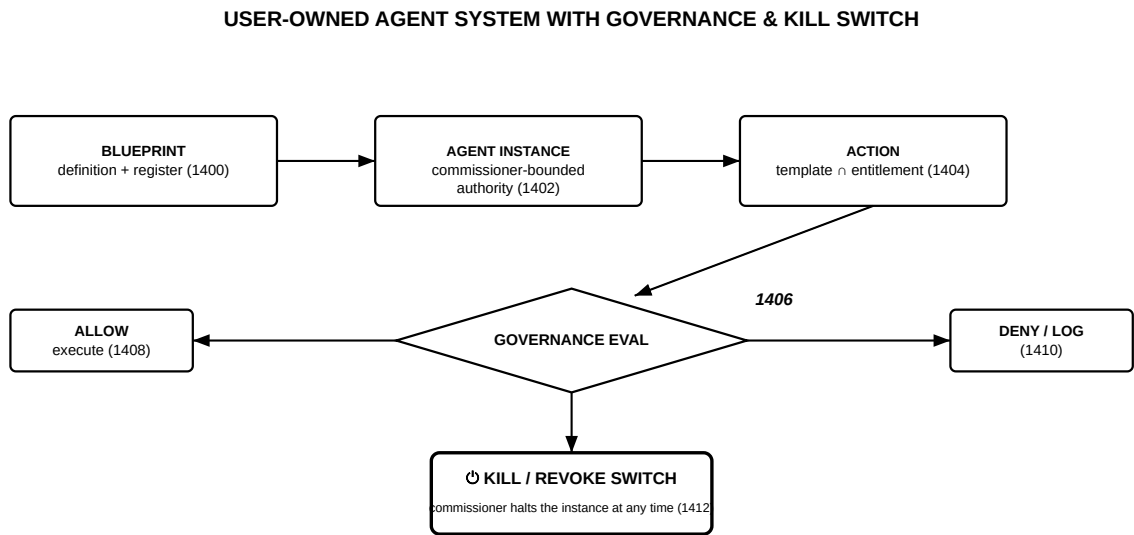


FIG. 21

FIG. 21 — Adaptive response engine — directive fixed before generation

FIG. 22

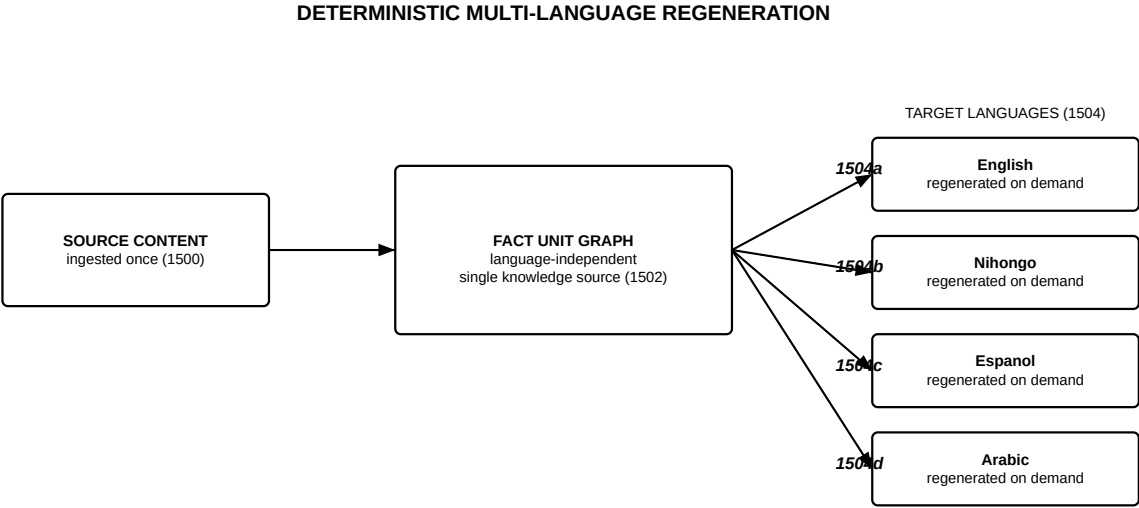


Governance is checked at every action; an agent can never exceed its commissioner, who holds an unconditional kill control.

FIG. 22

FIG. 22 — User-owned agent system with governance and kill switch

FIG. 23



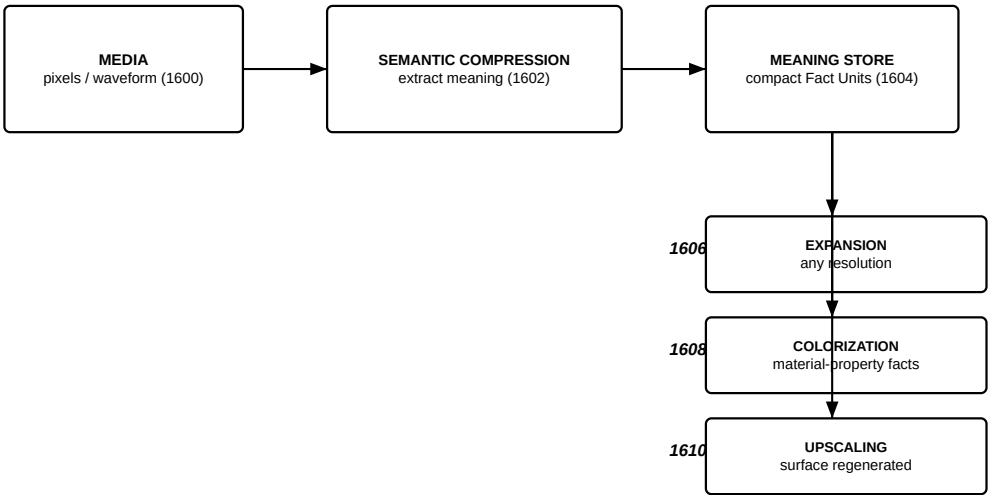
One graph, many surfaces: no per-language re-ingestion; each language is a projection of the same facts.

FIG. 23

FIG. 23 — Deterministic multi-language regeneration

FIG. 24

SEMANTIC COMPRESSION, EXPANSION, COLORIZATION & UPSCALING



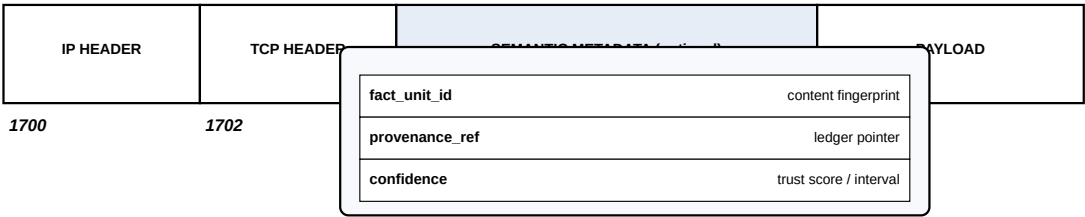
Store meaning, not surface; compression and regeneration are the same primitive — no generational quality loss.

FIG. 24

FIG. 24 — Semantic compression, expansion, colorization and upscaling

FIG. 25

QUERYBOOK-ENHANCED TCP/IP HYBRID PROTOCOL PACKET



A non-participating endpoint ignores the semantic header — provenance travels with the bits, backward-compatible.

FIG. 25

FIG. 25 — QueryBook-enhanced TCP/IP hybrid protocol packet

FIG. 26

NEUROMORPHIC COGNITIVE LAYER — ATTRACTOR CONVERGENCE

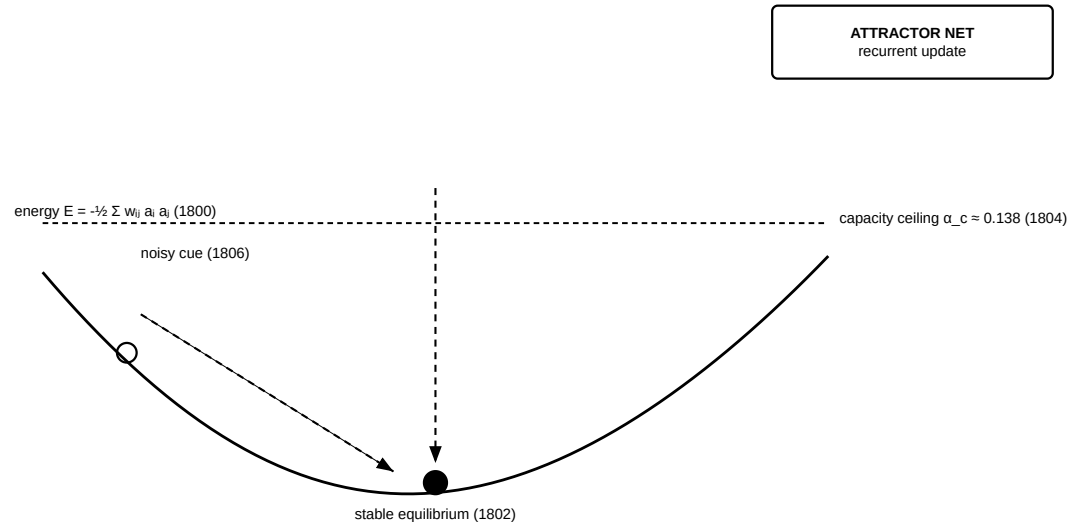
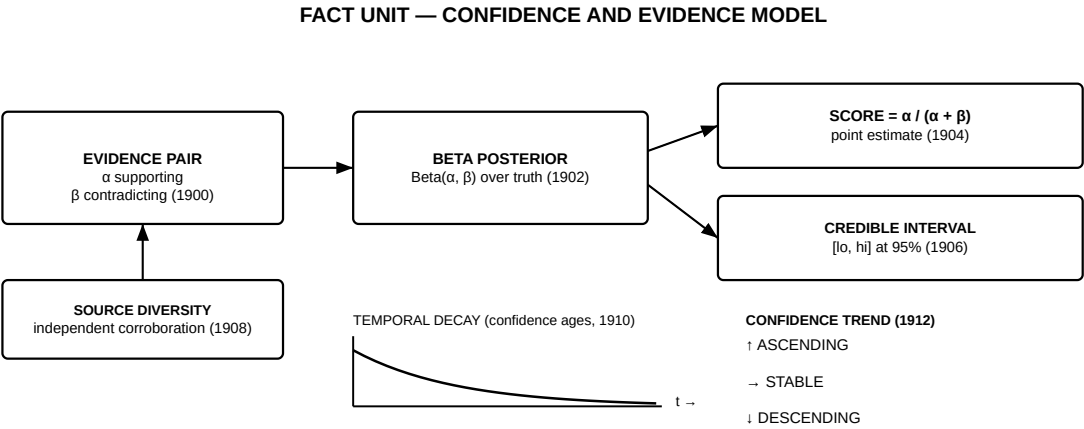


FIG. 26

FIG. 26 — Neuromorphic cognitive layer — attractor convergence

FIG. 27



Trust is a governed evidence estimate — not a single number: it carries an interval, ages over time, and reports its direction.

FIG. 27

FIG. 27 — Fact Unit — confidence and evidence model

FIG. 28

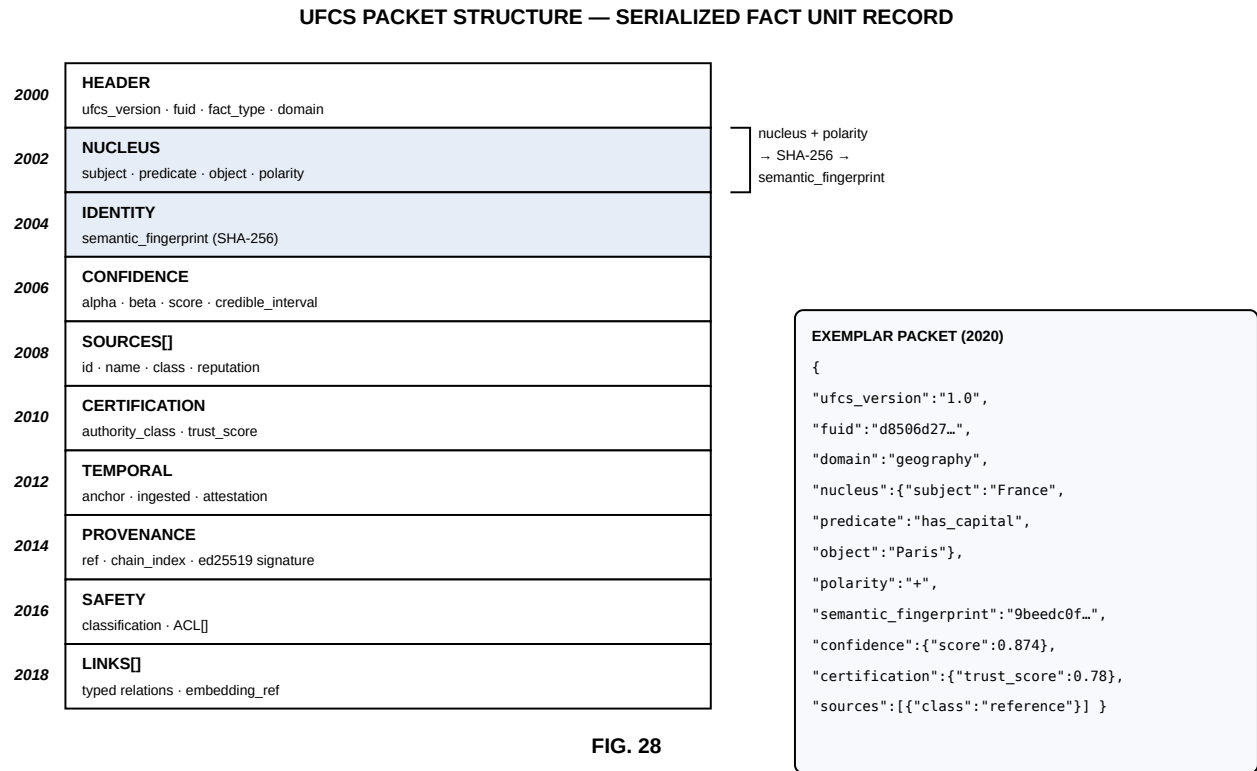


FIG. 28 — UFCS packet structure — serialized Fact Unit record

FIG. 29

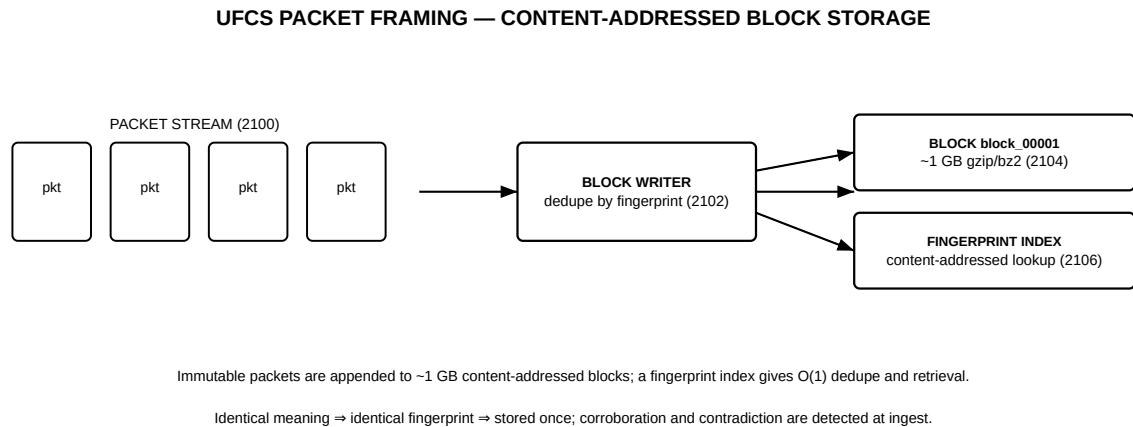


FIG. 29 — UFCS packet framing — content-addressed block storage

FIG. 30

FQL — QUERY ANATOMY AND WORKED EXAMPLE

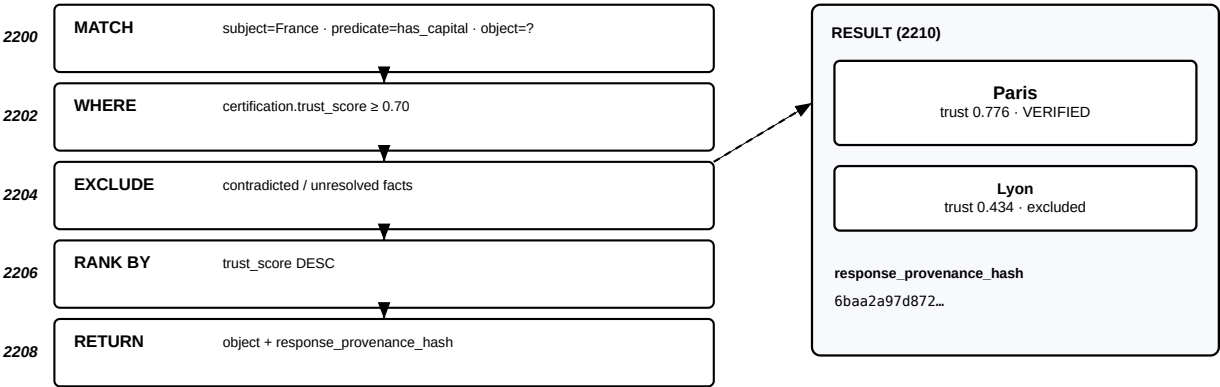


FIG. 30

FIG. 30 — FQL — query anatomy and worked example

FIG. 31

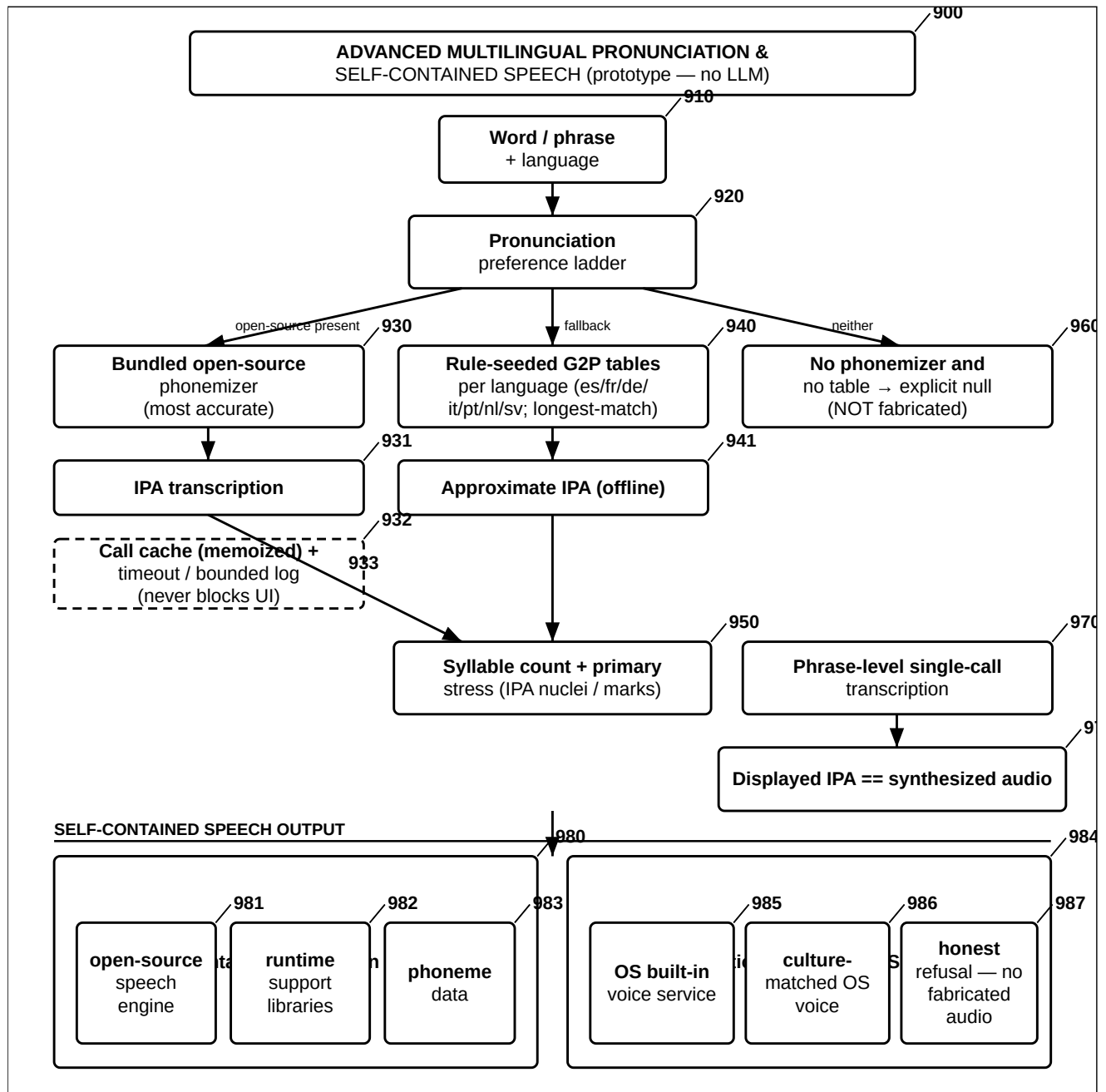


FIG. 31 — Advanced Multilingual Pronunciation and Self-Contained Speech (No LLM)

FIG. 32

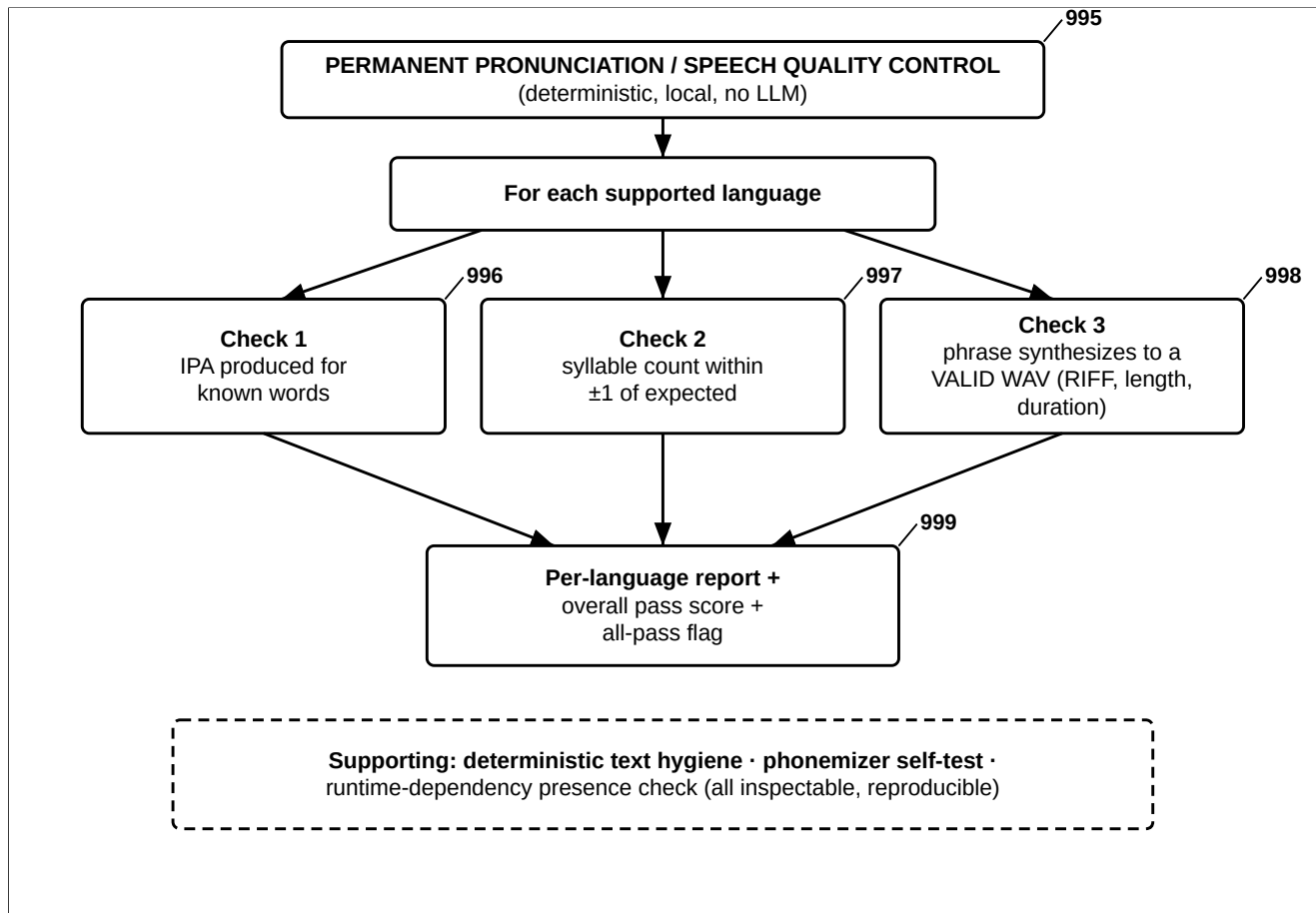


FIG. 32 — Permanent Pronunciation and Speech Quality-Control Battery (No LLM)

FIG. 33

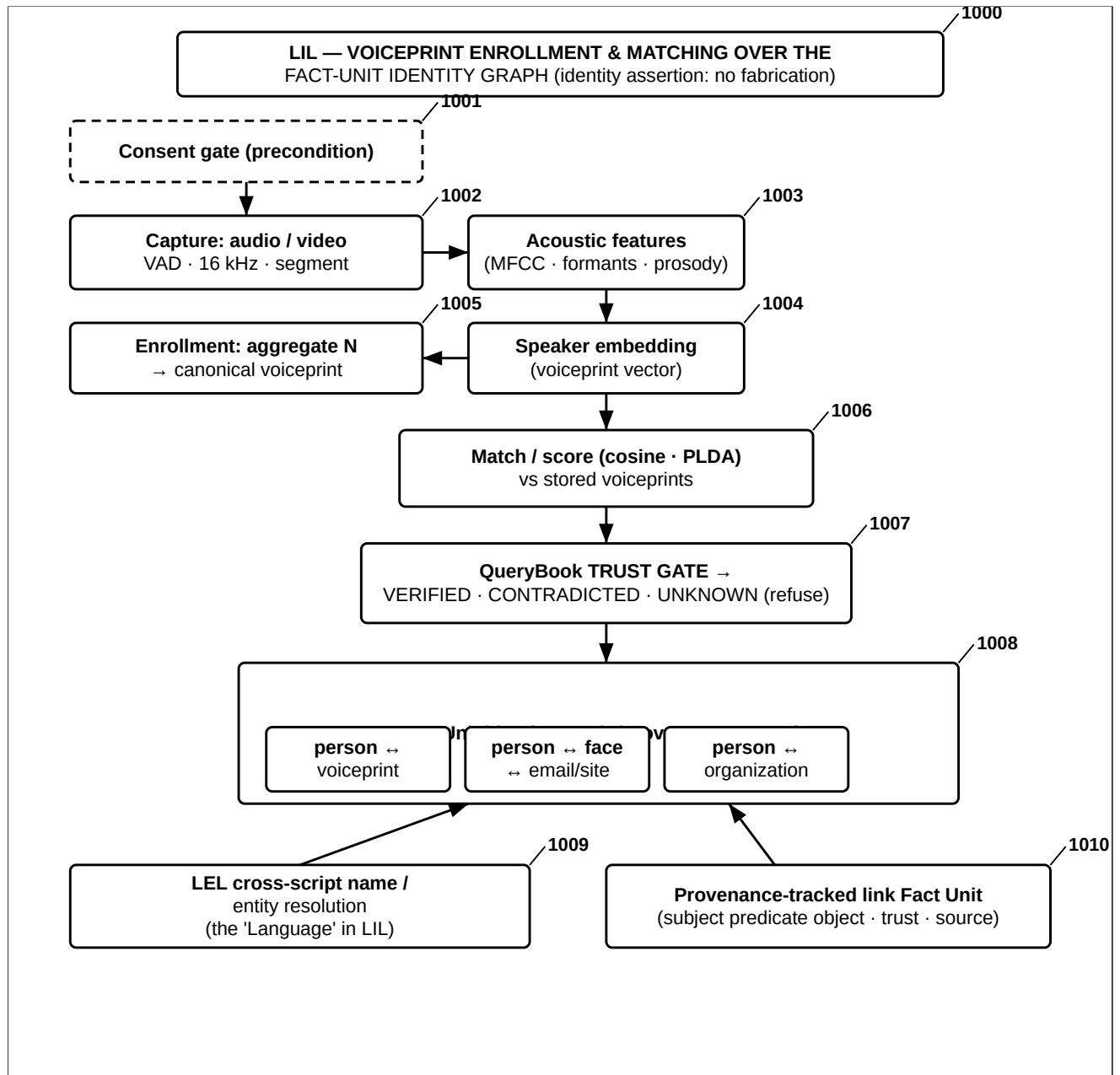


FIG. 33 — LIL — Voiceprint Enrollment and Matching over the Fact-Unit Identity Graph

FIG. 34

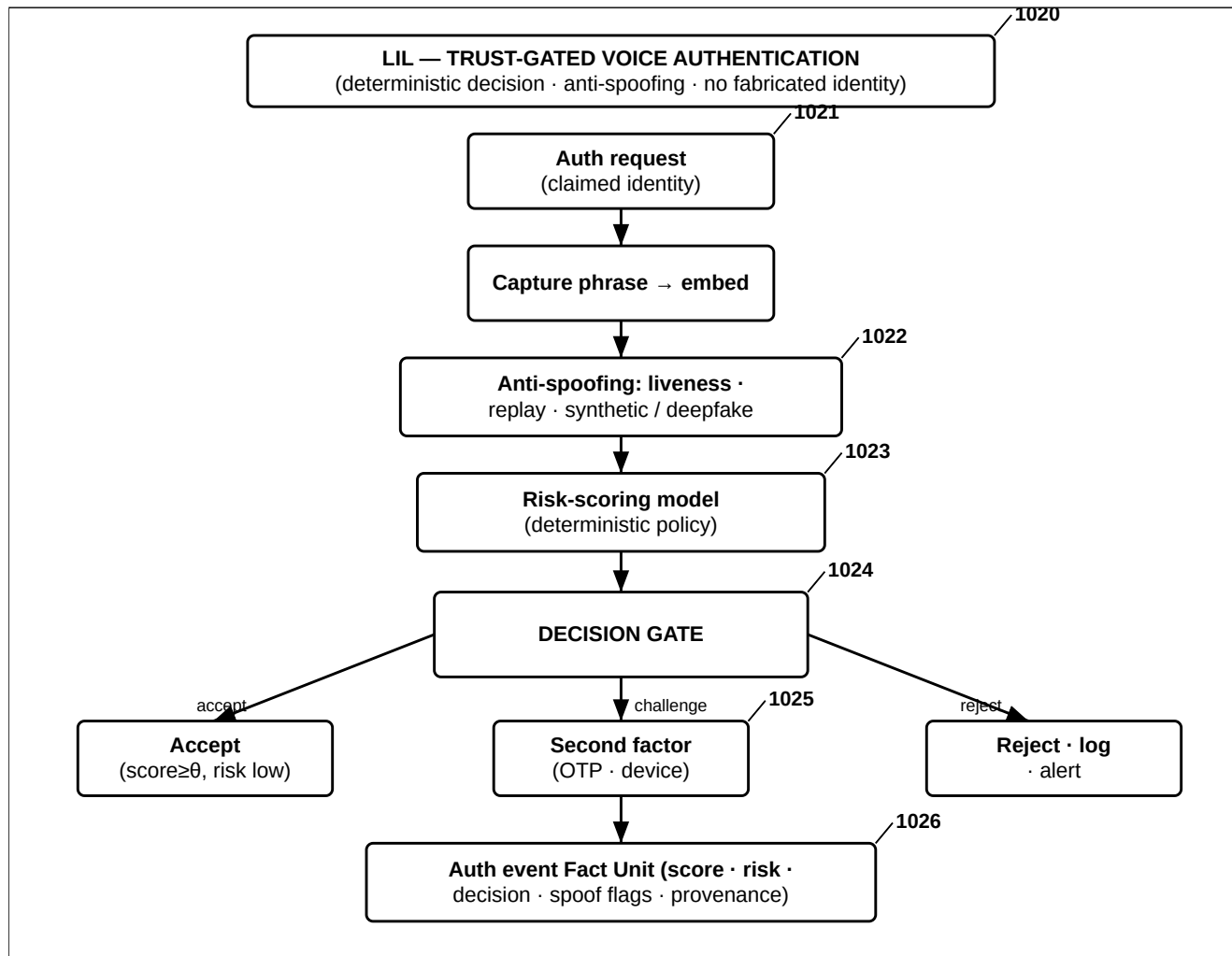


FIG. 34 — LIL — Trust-Gated Voice Authentication with Anti-Spoofing

DRAFT — NOT FILED — FOR REVIEW BY QUALIFIED PATENT COUNSEL PRIOR TO FILING. Companion drawings volume to the consolidated QueryBook Provisional Patent Application; figures and numerals are drawn from, and identical to, that application. Finalize to 37 CFR 1.84 before filing.